MILITARY SPECIFICATION

AN/APM-131 TEST HARDNESS RADAR

This specification be been approved by the Bureau of Naval Weapons, Department of the Navy

1.0 SCOPE

1.1 Scope - The kit covered by this specif ication shall be for flight line, shop use and shall provide the patch cables, or test adapters necessary to operate the specified components as part of the system, and provide necessary test points, when they are removed from their normal mounting position for servicing.

1.2 Classification " The AN/APM-131 Test Harness, Radar shall be of one type and shall Consist of the following:

Items	Quantity Rewired	Applicable Paragraph
Test Adapter	1	3.3.1.1
Test Adapter	1	3.3.1.2
Test Adapter	2	3.3.1.3
Test Adapter	2	3.3.1.4
Teat Adapter	2	3.3.1.5
Test Adapter	2	3.3.1.6
Test Adapter	2	3.3.1.7
Test Adapter	2	3.3.1.8
Test Adapter	1	3,3.1.9
Test Adapter	1	3.3.1.10
Test Adapter	1	3.3.1.11
Test Adapter	1	3.3.1.12
Test Adapter	1	3.3.1.13
Test Adapter	1	3.3.1.14
Test Adapter	1	3.3.1.15
Test Adapter	2	3.3.1.16
Test Adapter	1	3.3.1.17
Test Adapter	1	3.3.1.18
Test Adapter	1	3.3.1.19
Synchronizer Main Cable	1	3.3.2.1
Synchronizer Co=Ax Cable	4	3.3.2.2
Synchronizer Plug-in Unit Co-Ax Cables straps	1	3.3.2.3 3.3.2.4
Power Supply and Servo Plate Main Cable	1	3.3.2.5
Modulator Main Cable	2	3.3.2.6
Modulator Plate Lead	1	3.3.2.7
EFC Main Cable	1	3.3.2.8
EFC Co-Ax Cable	2	3,3.2.9
Indicator Low Voltage Unit Cable	1	3.3.2.10
Indicator Video Unit Main Cable	1	3.3.2.11
Indicator Video Unit Co-Ax Cable	2	3.3.2.12
Extension Mount ing	8	3.3.2.13
Extension Mounting	4	3.3.2.14
Ground Leads	6	3.3.2.15
Modulator Chassis Hooks and Line	1 set	3.3.3
Transit Case	1	3.3.4

2.0 APPLICABLE DOCUMENTS

2.1 General - The following documents of the Issue in effect on the date of Imitation for bids, form a part of this specif i cation, to the extent specified herein.

SPECIFICATION

Military

MIL-T-21200 Test Equipment for use with Electronic and Fire Control Systems; General specif ication for

FSC 4931

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SPECIFICATION (cont.)

MIL-N-18307	Nomenclature and Nameplates Airborne Electronic and Associated Equipment
MIL-P-17555	Electronic Miscellaneous Electrical Equipment (except Rotating Electrical Equipment) and Associated Repair Parts
MIL-T-18303	Test Procedures; Preproduction and Inspection, for Aircraft Electronic Equipment, Format for

2.2 Availability of Documents - When requesting specification, standards drawings and publication refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code CDS, 5801 Tabor Avenue, Philadelphia 20, Pennsylvania.

3.0 REQUIREMENTS

3.1 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In 80 doing the following shall govern:

(1) Parts and materials requirements shall conform to Specification MIL-T-21200 Parts selected from specifications called out in Specification MIL-T-21200 shall be limited to those values, styles, tolerance, etc., set forth in MIL-STD-242.

(2) Nonrepairable subassemblies, as outlined in Specification MIL-T-21200, shall be used when practicable. The general size of the subassembly, and the amount of Circuitry to be included therein shall be approved by the procuring activity. Nonrepairable subassemblies must be reliable. A ream-time-to-failure of more than 5000 hours should be the goal of each.

(3) When previously Produced models of this equipment did not use nonrepairable subassemblies, the design shall not be changed to employ nonrepairable assemblies without the approval of the procuring activity.

3.1.1 Nonstandard Parts and Material Approval - Approval for the use of nonstandard parts and materials shall be obtained as outlined in Specification MIL-T-21200.

3.1.2 Electron Devices - Electron tubes, transistors and diodes shall be chosen and applied, and the complements reported as outlined in Specification MIL-E-4682. The complement report must be submitted to the procuring activity for review and approval prior to preproduction testing.

3.2 Design and Construction

3.2.1 General - The equipment shall conform with all the applicable requirements of MIL-T-21200 for design, construction, and workmanship, unless otherwise specified herein.

3.2.2 Total Weight - The total weight of the equipment, Including the transit case, ehall be a minimum consistent with good design, and shall not exceed 180 pounds.

3.2.3 Operating Life

3.2.3.1 Reliable Operating Life - The equipment shall have a reliable operating life of at least 200 hours without removal for bench servicing. Parts requiring servicing or replacement at the end of this interval to renew this service shall be specified by the manufacturer.

3.2.3.2 Total Operating Life - The equipment shall have a minimum total operating life of 2000 hours with reasonable servicing and replacement of parts. Parts requiring replacement within this interval and the life of such parts shall be specified by the manufacturer.

3.2.4 Cabling and Connection - Cables, cabling, and grounding of cable shields shall be compatible with the requirement of the Radar System.

3.2.5 Nomenclature and Nameplates - Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with Specification MIL-N-18307.

3.2.6 Standard Conditions - The following conditions shall be used to establish normal performance characteristic under standard conditions and for making laboratory bench tests:

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Temperature	Room Ambient (30° C ±10° C)
Altitude	Normal Ground
Vibration	None
Humidity	Room Ambient up to 90% Relative Humidity

3.2.7 Semite Conditions - The equipment shall None satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified in MIL-T-21200 for Class 2B equipment.

3.2.7.1 Moisture and Fungus Resistant Treatments - The equipment shall be fungus-proofed by selection of parts, and materials shall be treated prior to their use in the equipment.

3.3 Detail Design - The Test Harness, Radar AN/APM-131 shall contain the following items to service the AN/APQ-72 Radar.

3.3.1 Test Adapters - General - The following teat adapters shall be provided for readily access ible testing of the AN/APQ-72 plug-in units specified. Test points shall be provided on each test adapter for all pin connection. The test points shall be located on the adapter 60 as to be readily accessible in use.

3.3.1.1 Test Adapter - One Required - This test adapter shall provide means for servicing plug-in unit A3001 on the Synchronizer, Electrical SN-233/APQ-72.

3.3.1.2 Test Adapter - One Required - This test adapter shall provide means for servicing plug-in units A3002, A3003 and A3201 on the Synchronizer, Electrical SN-233/MQ-72.

3.3.1.3 Test Adapter - Two Required - These test adapters shall provide means for servicing relays K2201, K2202, K2203, K2204 and K2207 on the Synchronizer, Electrical SN-233/APQ-72; K4301 on the Control, Radar Set C -2452/APQ-72; K4803, K4804, K4805 and K4808 on the Servo Plate Assembly of the Control, Power Supply C -2783 /APQ-72; K4901, K4902, K4904 and K4905 on the Search Command Chassis of the Control, Power Supply C-2783 /APQ-72 and K3701, K3708 and K3711 on the Control, Indi - cator C-2451/APQ-72.

3.3.1.4 Test Adapter - Two Required - These test adapters l shall provide means servicing relays K2205, K2206 K2209, K2211, K2212 and K2218 on the Synchronizer, Electrical SN-233 APQ-72; K100l and K1002 on the Chassis Assembly of the Electrical Frequency Control of the Radar Set Group OA-l849/APQ-72 and K4604 on the Control, Power Supply C-2783/APQ-72.

3.3.1.5 Test Adapter - Two Required - These adapters shall provide means for servicing relays K2208, K2213, K2214, K2215, K2216, K2217, K2219, K2220, K2221, K2222, K2223, K2224 and K2226 on the Synchronizer, Electrical SN-233/APQ-72 and K4302, K4303 and K4305 on the Control, Radar Set C-2 452/APQ-72.

3.3.1.6 Test Adapter - Two Required - These test adapters shall provide means for servicing relays K2210, on the Synchronizer, Electrical SN-233/APQ-72; K3702, K3703, K3704, K3706, K3707, K3709 and K3710 on the Control, Indicator C-2451/APQ-72; K1003 on the Chassis Assembly of the Electrical Frequency Control of the Radar Set Croup QA-1849/APQ-72; K4806, K4807 and K4903 on the Servo Plate Assembly of the Control, Power Supply C -2783/APQ-72; K4801 and K4802 on the Track Demodulator Chassis of the Control, Power Supply C-2783/APQ-72; K4907 on the Search Command Chassis of the Control Power Supply C-2783/APQ-72 and K4605 on the Control, Power Supply C-2783/APQ-72.

3.3.1.7 Test Adapter - Two Required - These test adapters shall provide means for servicing units Y2201, Y2202 and Y2203 on the Synchronizer, Electrical 8N-233/APQ-72; V4705, V4706, V4707 on the Control, Power Supply C -2783/APQ-72; V4805 and V4806 on the Track Demodulator Chassis of the Control, Power Supply C -2783 /APQ-72; and G1001 on the Chassis Assembly of the Frequency Control of the Radar Set Group OA-1849 /APQ-72.

3.3.1.8 Test Adapter - Two Required - These test adapters shall provide means for servicing plug-in units A2901, A2902, A2903, Z2905, Z2906 Z2907 Z2908, Z3101, Z3102, Z3103 Z3104, Z3105 and Z3106 on the Synchronizer, Electrical SN-233/APQ-72; and A1001, A1002, A1003, A1004 and A1005 on the Chassis Assembly of the Electrical Frequency Control of the Radar Set Group OA-1849/APQ-72.

3.3.1.9 Test Adapter - One Required - This test adapter shall provide means for servicing plug-in units A2904, Z2910, Z2909, Z3107 and Z3108 on the Synchronizer, Electrical SN-233/APQ-72.

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3.3.1.10 Test Adapter - One Required - This test adapter shall provide means for servicing plug-in units A4601 and A4b03 on the Control, Power Supply C -2783 /APQ-72 and A4801 on the Servo Plate Assembly of the Control, Power Supply C-2783 /APQ-72,

3.3.1.11 Test Adapter - One Required - This test adapter shall provide means for servicing plug-in Units A4602, A4604, A4605, A4701, A4702 and A4703 on the Control, Power Supply C-2783/APQ-72; and A4802 and A4803 on the Servo Plate Assembly of the Control, Power Supply c-2783/ APQ-72.

3.3.1.12 Test Adapter - This test adapter shall provide means for servicing relay K4603 on tbe Control, Power Supply c-2783/APQ-72.

3.3.1.13 Test Adapter - One Required - This test adapter shall provide means for servicing tubes V4601, V4602, V4603, V4604, V4605, V4606, V4701, V4702, V4703 and V4704 on the Control, Power Supply c-2783 /APQ-72.

3.3.1.14 Test Adapter - One Required - This test adapter shall provide means for servicing the tuning fork A4704 (and V4901 and V4904) on the Control, Power Supply C-2783/APQ-72.

3.3.1.15 Test Adapter - One Required - This test adapter shall provide meam for servicing relay K704 on Radar Set Group QA-1849/APQ-72.

3.3.1.16 Test Adapter - Two Required - These test adapters shall provide means for servicing plug-in units A3701, A3702, A3703, A3704, A3705, A3707, A3708, A3709, A3710, A3711.) A3713, A3714, A3715, A3716, A3717 and A3719 on the Control, indicator C-2451/APQ-72.

3.3.1.17 Test Adapter - One Required - This test adapter shall provide means for servicing plug-in units A3706, A3712, A3718 and A3720 on the Control, Indicator C-2451/APQ-72.

3.3,1.18 Test Adapter - One Required - This test adapter shall provide means for servicing plug-in unit A401 and A402 on the Radar Set Group OA1849/APQ-72.

3.3.1.19 Test Adapter - One Required - This test adapter shall provide means for servicing relays K404, K405, K406 and K407 on the Radar Set Group OA1849/APQ-72.

3.3.2 Patch Cables - Patch cables shall be provided as follows:

3.3.2.1 Synchronizer Main Cable - One Required - This cable, when used in conjunction with the Synchronizer Co-ax Cables (paragraph 3.3.2.2), shall permit normal operation within the system of the Synchronizer, Electrical SN-233/APQ-72 when it is removed from the system to a test bench located adjacent to the system. The cable shall be 3 feet ±2 inches in length. All blank pins are to be wired pin to pin with #22 low voltage general hook-up wire.

3.3.2.2 Synchronizer Co-ax Cable - Four Required - These cables, when used in con-Junction with the Synchronizer Main Cable (paragraph 3.3.2.1), shall permit normal operation within tbe system of the Synchronizer, Electrical SN-233/APQ-72 when Synchronizer is removed from system to a test bench adjacent to the system. The cables shall be 4 feet ±2 inches in length and shall be made of 95-ohm subminiature co-ax cable. The connectors shall be compatible with PJ2202, PJ2203, PJ2204 and PJ2205 of the Synchronizer, Electrical SN-233/APQ-72.

3 3.2.3 Synchronizer Plug-in Unit Co-Ax Cable - One Required - This cable sball extend the connection from the chassis to the plug-in unit of the Synchronizer, Electrical SN-233/ APQ-72 when the plug-in unit is mounted on a Test Adapter (paragraph 3.3.1.1 and 3,3.1.2). This cable shall be 8 inches ±0.25 inch in length, and made of RG59/U cable. The connectors shall be compatible with PJ3202, PJ3001 and PJ3204 of the Synchronizer, Electrical SN-233/APQ-72.

3.3.2.4 straps - These straps shall be used in lieu of straps per WEC Dwg. 401R243 G01-G06 an required when plug-in unit(s) A3201, A3003 is raised on Teat Adapter (paragraph 3.3.1.1 and paragraph 3.3.2.2).

3.3.2.5 Power Supply and Servo Plate Main Cable - This cable shall permit normal operation within the system of the Power Supply and Servo Plate C-2783 /APQ-72 when It is removed from the system to a test bench located adjacent to the system. The cable sball consist of three individual cables 42 inches ±2.5 inches in length. The individual cable shall be formed into one main cable leaving the individual cables free at both ends for a length of 6 inches ±.25 Inch.

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3.3.2.6 Modulator Main Cable - Two Required - These cables when used in conjunction with the Modulator Plate lead (paragraph 3.3.2,7), shall permit normal operation of the Modulator Control Chassis when it is removed from the Radar Set Group OA-1849/APQ-72. This cable shall be 9 inches ±.25 inch in length. The wire shall be #20 unshielded low voltage general hook-up wire, wired pin to pin, utilizing the hooks and line of paragraph 3.3.3.

3.3.2.7 Modulator Plate Lead - One Required - This lead, when used in conjunction with the Modulator Main Cables (paragraph 3.3.2.6), shall permit normal operation of the Modulator Control Chassis when it is removed from the Radar Set Group OA-1849/APQ-72 utilizing the hooks and line of paragraph 3.3.3 to the system. This lead shall be 16 inches ±.25 inch in length.

3.3.2.8 EFC Main Cable - One Required - This cable, when used in conjunction with the EFC Co-ax Cables (Paragraph 3.3 .2.9), shall permit normal operation within the system of the Electrical Frequency Control of the Radar Set Group OA-1849/APQ-72 when it is removed from the syetem to a bench adjacent to the system. The cable shall be 3 feet ±1 inch In length.

3.3.2.9 EFC Co-Ax - Two Required - These cables, when used in conjunction with the EFC Main Cable (paragraph 3.3.2.8) shall permit normal operation within the system of the Electrical Frequency Control of the Radar Set Group OA-1849/APQ-72 when it is removed from the system to a test bench adjacent to the system. These cables shall be 3 feet ±1 inch In length.

3.3.2.10 Indicator Low Voltage Unit Cable - One Required - This cable shall permit normal operation within the system of the Indicator Low Voltage Unit when it is removed from the Indicator, Azimuth - Elevation - Range IP-447/APQ-72 or Indicator, Pilot's Azimuth - Elevation - Range IP/APQ-72 to a test bench located adjacent to the system. This cable shall be 18 inches ±1 inch in length.

3.3.2.11 Indicator Video Unit Main Cable - One Required - This cable, when used in conjunction with the Indicator Video Unit Co-ax Cables (paragraph 3.3.2.12), shall permit normal operation within the system of the Indicator Video Unit when it is removed from the Indicator, Azimuth - Elevation - Range IP-447/APQ-72 or Indicator, pilot's, Azimuth - Elevation - Range IP/APQ-72 to a test bench located adjacent to the system. This cable shall be 18 inches ±1 inch in length.

3.3.2.12 Indicator Video Unit Co-ax Cable - Two Required - These cables, when used in conjunction with the Indicator Video Unit Main Cable (paragraph 3.3.2.ll), shall permit normal operation within the system of the Indicator Video Unit when it is removed from the Indicator, Azimuth Elevation - Range IP-447/APQ-72 or Indicator, Pilot's, Azimuth, Elevation - Range IP/APQ-72 to a best bench located adjacent to the system. These cables shall be 18 inches ±1 inch in length.

3.3.2.13 Mtg. extensions used in conjunction with molded unit adapters to assure positive mtg. action for unite requiring size 6-32 THD.

3.3.2.14 Mtg. extensions used In conjunction with molded unit adapters to assure positive mtg. action for units requiring size 4-40 THD.

3.3.2.15 Ground Leads (405R093G01 through 6) used in conjunction with the various molded unit and relay adapters to provide positive ground connection to chassis.

3.3.3 Modulator Chassis - Hook and Line - Shall consist of four supporting metal hooks and one nonconducting line to support modulator chassis for servicing.

3.3.4 Transit Case - The transit ease shall be designed to hold all the items included in the Test Harness, Radar (paragraph 3.3.1 to paragraph 3.3.3). Each item shall have a particular location within the tranait case.

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4. QUALITY ASSURANCE PROVISIONS

4.1 General - Items covered by this specification shall be subjected to the following tests to determine compliance all applicable requirements:

(1) Preproduction Tests
(2) Production Sample Tests
(3) Acceptance Tests

4.1.1 Inspection - The supplier is responsible for the performance of all inspection requirements as specified herein. Except as specified, the supplier may utilize his own on any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specifications where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preproduction Tests - Preproduction tests shall be made on an equipment representative of the production equipments to be supplied under the contract. Preproduction tests shall consist of the following:

(1) Contractor's Demonstration Tests
(2) Service Approval Tests

4.2.1 Contractor's Demonstration Tests - Contractor's demonstration teets shall be accomplished under the responaibility of the contractor and shall be conducted in accordance with the approved test procedure of 4.5. The government inspector end the procuring activity shall be advised when tests are to be conducted 50 that a representative may be designated to witness or supervise the tests when so desired. Contractor not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory.

4.2.1.1 The contractor shall submit all data collected in conducting these tests to the procuring agency for review and approval. This data shall include a list of all electrical and electronic parts giving tbeir specified voltage, current and temperature rating and the applied circuit voltage, current and ambient and surface temperatures. The ambient and surface temperature shall be obtained under the extreme high temperature operating condition. This date may be included as part of the preliminary parts list.

4.2.2 Service Approval Tests - At the completion of the contractor's demonstration tests and when required by the procuring activity, the equipment shall be delivered to a specified government laboratory for additional testing. This additional testing may consist of duplicating tests perviously conducted and such Other tests as are deemed necessary to determine compliance with all applicable design and performance requirements.

4.2.2.1 Accessory Material - In addition to the complete equipment submitted for service approval te sts, the contractor shall also submit the accessory material and design and teat data specifiedin Specification MIL-T-21200. This information shall indicate the physical and electrical characteristics of the equipment and establish the equipment's compliance with applicable requirements.

4.2.3 Scope of Tests - Preproduction tests shall include all tests deemed necessary to determine that the equipment meets all the requirements of this specification and the contract. Preproduction tests shall include environmental tests in accordance with Specification MIL-T-5422, including the Sand and Dust Test and modifying the test limits of the applicable tests to correspond with the limits specified herein Interference tests shall be conducted in accordance with Specification MIL-X-6181. Drop tests for transit and combination cases shall be In accordance with Specification MIL-T-21200.

4.2.4 Preproduction Approval - Approval of the preproduction model shall be by the procuring l ctivity upon satisfactory completion of all tests. No production equipment shall be delivered prior to the approval of the preproduction model. Pre-fabrication of production equipment prior to the approval of the preproduction model is at the contractor's own risk. The approved preproduction model will be returned to the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction model shall not be considered as one of the equipments under the contract; however, it may be reworked by the contractor and submitted for acceptance as a production equipment.

4.2.5 Equipments supplied under the contract shall in all respects, including design,

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construction, workmanship, performance and quality be equivalent to the lpproved preproduction sample. Each equipment shall be capable of successfully passing the same tests imposed on the preproduction sample. Evidence of non-compliance with the above shall constitute cause f or rejection or if already accepted by the government It shall be tbe obligation of the contractor to makes necessary corrections l s approved by the procuring activity.

4.3 Production Sample Tests - When requested by the procuring activity, one equipment shall be selected from the first 10 production equipments submitted for acceptance and sent to a designated government laboratory for tests. Tbe equipment shall be eel-acted by the government inspection after the equipment baa successfully passed all individual tests.

4.3.1 Scope of Teats - This equipment may be subjected to any and all tests the procuring l ctitit y deems necessary to a ssure that the production equipment is equivalent to the approved pre-production sample in design, construction, workmanship, performance and quality and that it is capable of meeting all applicable requirement.

4.3.2 Production Sample Approval - Approval of the production sample shall be by the procuring activity upon satisfactory completion of all teats. Any design, material or performance defect made evident during this test shall be corrected by tbe contractor to the satisfaction of the procuring activity. Upon failure of the production sample to pass any of the tests delivery of equipment under the contract shall cease until proper corrective action is approved and accomplished. Corrective l ction shall also be accomplished on equipment previously accepted when requested by the procuring activity.

4.3.3 Reconditioning of Production Test Sample - On completion of the production sample teat the equipment shall be reworked by the contractor by replacing all worn or damaged items. After reworking the contractor shall resubmit the equipment for acceptance.

4.4 Acceptsnce Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of tbe government Inspector. Contractors not having testing facilities satisfactory to the procuring activity shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall f urnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of maerial during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished products. Acceptance tests sball consist of the following:

(1) Individual Tests

(2) Sampling Tests

(3) Special Tests

4.4.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual teats. These tests shall be adequate to determine compliance witb the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(1) Examination of Product

(2) Operational Test

(3) Manufacturing Reliability Test

4.4.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

4.4.1.2 Operational Test - Each equipment shall be operated ling enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to l aoure satisfactory equipment operation.

4.4.1.3 Manufacturing Reliability Test - Each equipment shall be operated under the conditions specified herein for a period of 6 hours without failure. A failure shall be defined as anything which causes malfunctioning of the equipment. Only those adjustments will be permitted which can be made by using such controls end adjustments that are accessible to the operator during the normal use of the equipment.

Temperature	Ambient room

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Humidity	Ambient room
Vibration	Any selected frequency within the range of 20 to 30 cps (excluding resonant points) and a minimum amplitude of ±3 g's

The equipment shall be vibrated (without vibration isolators) for a period of 10 minutes prior to the beginning of the 6-hour period of operation. Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plane for 5 minutes and lateral to that plane for 5 minutes. Where it is not feasible to vibrate the equipment in 2 directions the vertical direction shall be used. During the 6-hour period of operation following, the 10-minute vibration period, the equipment shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, it should be repaired and the test started over, except that the 10-minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures. The 6-hour period specified above may be composed of two 3-hour periods to conform with standard working hours,

4.4.2 Sampling Tests - Sampling tests shall be conducted on equipments which have successfully passed the individual tests. Equipments shall be selected for sampling tests by the Government Inspector in accordance with the following:

Quantity of Equipments Offered for Acceptance	Quantity to be Selected For Sampling Test
First 10	1
Next 50	1
Next 75	1
Next 100	1
	1 for each additional 200 or fraction thereof

4.4.2.1 Scope of Tests - As a minimum, each equipment selected for sampling tests shall be subjected to the following tests:

(1) Complete operational test at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

(2) Operational test at certain environmental conditions. The conditions may vary for each equipment tested and should be based on results of the preproduction, production and special tests.

4.4.3 Special Tests - Special tests shall be conducted on a quantity of equipments for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling tests.

4.4.3.1 Special Tests Schedule - Selection of equipments for special tests shall be made as follows; except that when a Production Sample is requested and delivered to a government laboratory for test (See para. 4.3) the requirement for a special test on an early production equipment may be omitted.

(1) On an early production equipment.

(2) On an early equipment after an engineering change.

(3) Whenever failure reports or other information indicate additional equipments should be tested. (This will be determined by the procuring activity.)

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4.4.3.2 Scope of Tests - Special tests shall consist of such tests as are approved by the procuring activity. Tests procedures previously approved for the preproduction and sampling tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests. As a minimum, the following tests shall be made, except for those equipments selected as a result of design changes and failure reports, only those tests required to check the characteristics in question need be conducted.

(1) Temperature - Altitude Tests

(2) Vibration Test (60 minutes cycling in each of 3 planes)

(3) Humidity Tests

4.4.4 Equipment Failure - Should a failure occur during either the sampling or special test, the following action shall be taken:

(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case or design defect.

(3) Submit to the procuring activity for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occuring in future tests.

(4) Where practical, include a test in the individual test to check all equipment for this requirement to assure that the defect has been satisfactorily corrected.

(5) Whenever a failure occurs, shift to one (1) sample unit out of ten (10) and proceed with testing as indicated in paragraph 4.4.2

4.5 Test Procedures - The procedures used for conducting preproduction tests and acceptance tests shall be prepared by the contractor. The right is reserved by the procuring activity or the government inspector to modify or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL-T-18303 shall be used as a guide for preparation of test procedures. When test procedures are available from previous contracts such procedures may be used when approved by the procuring activity.

4.6 Presubmission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.7 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting , full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the government inspector.

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5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed, and marked for the level of shipment specified in the contract or order in accordance with Specification MIL-P-17555.

6. NOTES

6.1 Use - The Test Harness provides patch cables and test adapters required to operate components of the Radar System, AN/APQ-72, as part of the system while providing test points, when the components are removed from their normal mounting position for servicing.

6.2 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

6.3 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. There it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in Specification MIL-T-21200, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accomplished by complete supporting information.

6.4 The parentheses (*), when used in the type designation, shall be replaced by either a number or letter furnished by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with 3.2.5. The complete type number shall be used on nameplates, shipping records and instruction books, as applicable.

NOTICE - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished or in any way supplied the said drawings, specifications or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

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